TrendPilot

A Local Agentic AI Assistant Using Ollama

AIRI Team PITB - AI Internship Task 2: End-to-End Agentic AI Project

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1. Project Overview

I built **TrendPilot**, a local agentic AI assistant that turns a topic into a complete social media content plan. The user enters a topic, platform, tone and output type. The agent writes a plan, calls tools to generate content angles, a post, hashtags and an optional reel script, reviews its own work, rewrites the post when the review score is low, saves the result to Markdown files and remembers the run in a JSON memory file.

The whole system runs offline on a CPU-only laptop using **Ollama** and **Gemma 3:4B**. No cloud API is used. The goal was to understand how an agentic workflow differs from a simple chatbot.

2. Tools and Technologies Used

- Python 3.13, VS Code, virtual environment
- Ollama with **gemma3:4b** (3.4 GB, running 100% on CPU, context 4096)
- requests library for the Ollama API (custom Python agent, no LangChain)
- JSON memory (memory.json) and Markdown file saving
- Git and GitHub for version control and submission

3. Agent Workflow

The agent follows a fixed pipeline and prints `[Tool Called]: ...` for every tool it uses:

Step	What the agent does
1. User input	Topic, platform, tone and output type
2. Planning	Agent writes a numbered plan; memory shows the previous topic
3. Content angles	Trend Idea Generator creates 3 angles and picks the best one
4. Generation	Caption Writer, Hashtag Generator, and Reel Script Generator (if requested)
5. Review	Content Reviewer gives a score out of 10 and improvement tips
6. Self-improvement	If the score is below 8, the post is rewritten and reviewed again
7. Saving	File Saver writes the result into 3 output folders; Memory Tool stores the run

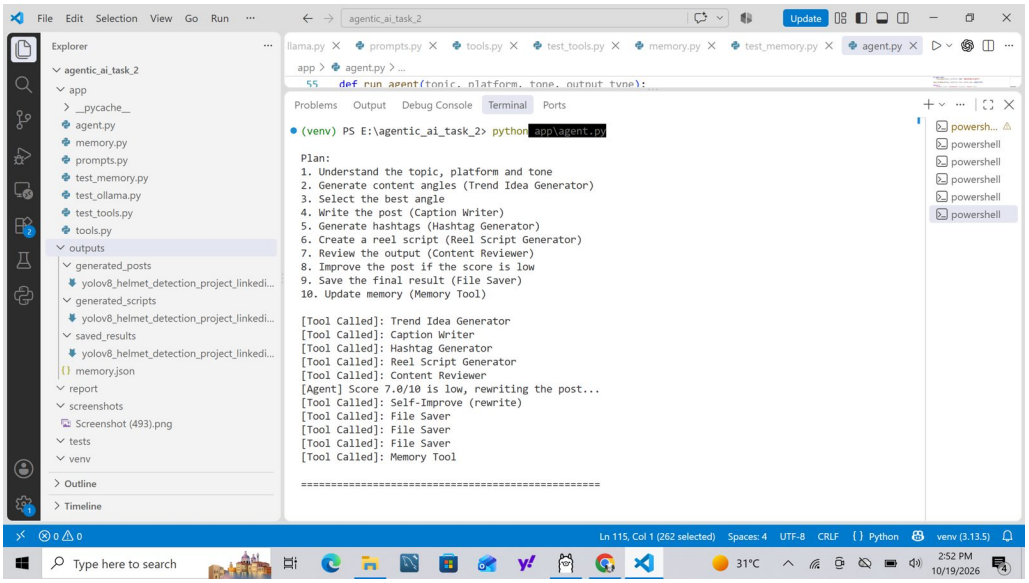


Figure 1: Agent plan and [Tool Called] log for the YOLOv8 helmet detection run

4. Tools Implemented

Tool	Description
Trend Idea Generator	Generates 3 content angles for the topic and platform, then selects the best one.
Caption Writer	Writes a platform-specific post following length and style rules (LinkedIn, Instagram, X/Twitter, YouTube Shorts).
Hashtag Generator	Produces 6 relevant one-word hashtags that start with #.
Reel Script Generator	Creates a 30-second script with hook, 3 scenes and a closing line.
Content Reviewer	Scores the post out of 10 and gives 3 improvement tips.
File Saver (mandatory)	Saves results as Markdown files in generated_posts, generated_scripts and saved_results.
Memory Tool	Stores the previous topic, platform, tone and saved file path in memory.json.

5. Demo and Outputs

The demo runs from the terminal. The user enters a topic, platform, tone and output type, and the agent prints the final post, hashtags, reel script and review, then saves everything. Example input: *YOLOv8 helmet detection project | LinkedIn | Professional but exciting | Post + Reel Script*.

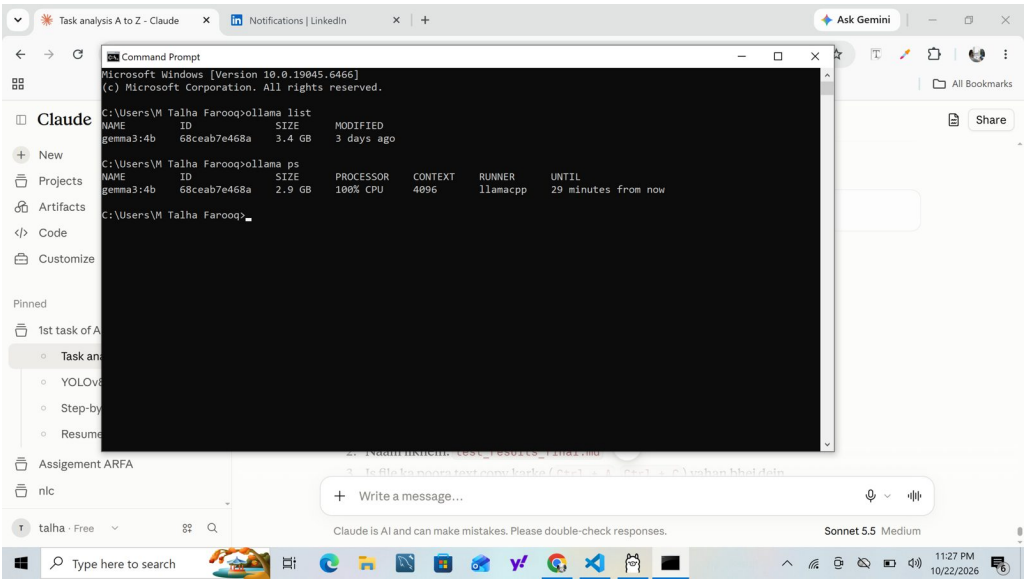


Figure 2: ollama list and ollama ps - gemma3:4b running locally on 100% CPU

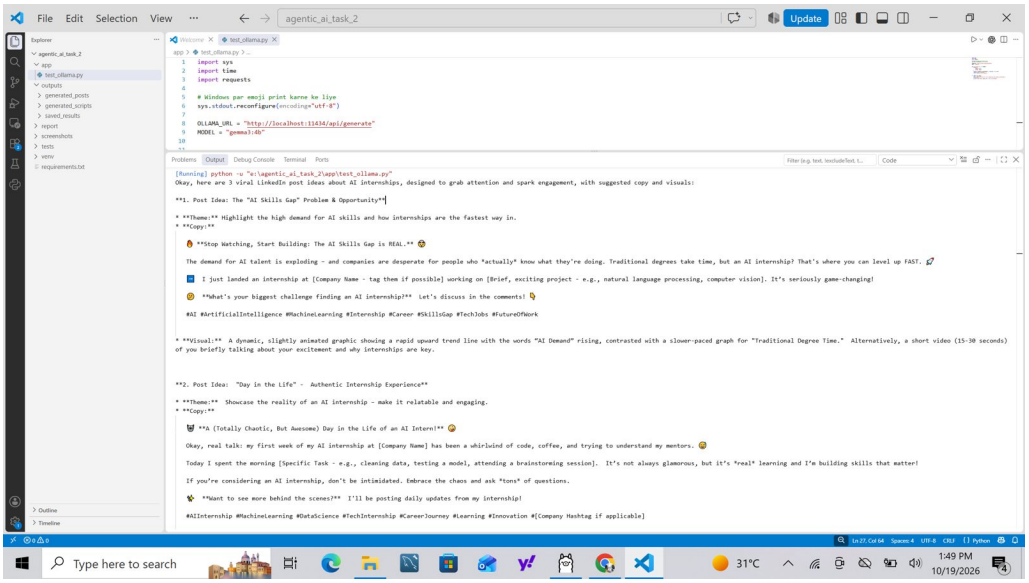


Figure 3: First Python connection to Ollama (AI internship post ideas)

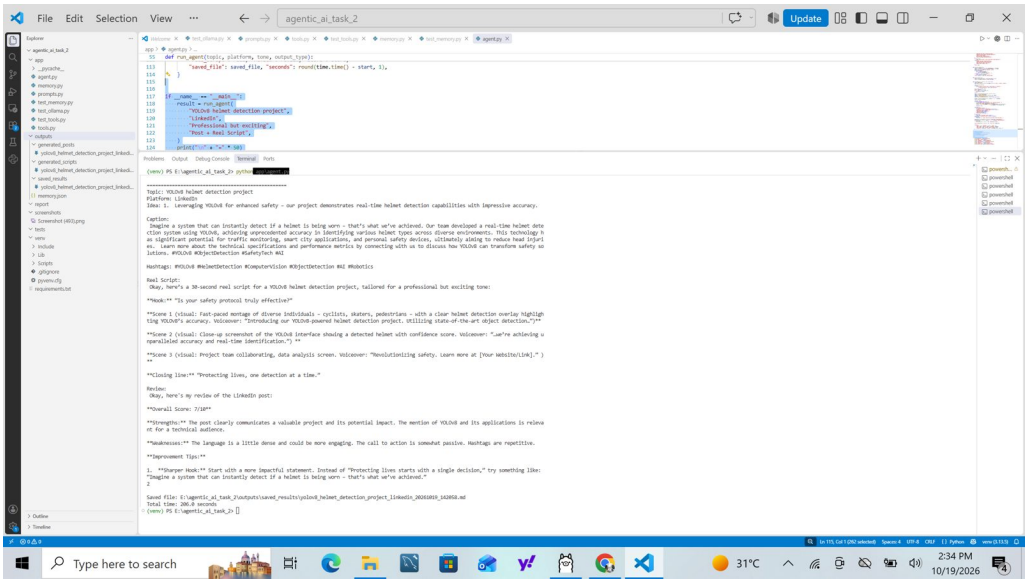


Figure 4: Final output of the first version of the agent (19 Oct)

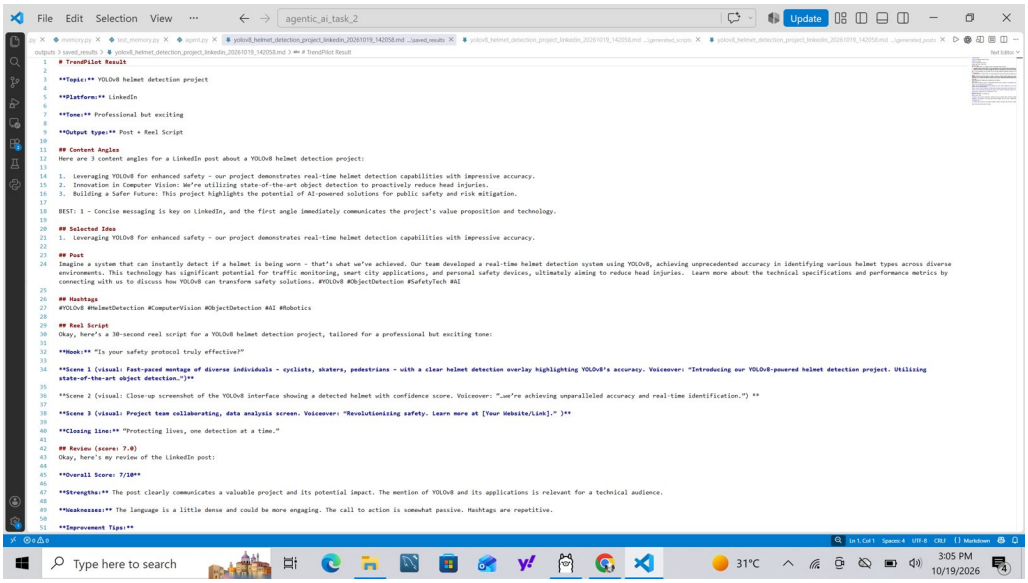


Figure 5: Saved Markdown result file in outputs/saved_results

Note: Figures 4 and 5 are from the first version. The post contains a placeholder (*[Your Website/Link]*) and exaggerated wording such as 'unparalleled accuracy'. These problems were fixed in later versions (see Error Analysis).

6. Testing

I tested the agent on 10 different inputs covering technical, motivational and humorous topics on LinkedIn, Instagram, X/Twitter and YouTube Shorts. The table shows the final test run (version 2, after prompt fixes). All 10 tests finished without errors and passed the automatic rule checks. Total time was 1824.5 seconds (30.4 minutes), about 3 minutes per run on CPU.

#	Topic	Platform	Tone	Output	Score before / after rewrite	Time (s)	Useful
1	YOLOv8 helmet detection	LinkedIn	Professional but exciting	Post	7 / 7	226.8	Partly
2	AI internship experience	LinkedIn	Motivational	Post	7 / 7	208.6	Partly
3	Python learning journey	Instagram	Funny	Reel script	7 / 7	198.0	No
4	Machine learning project	X/Twitter	Short and catchy	Post	7 / 6	123.3	Partly
5	Data annotation struggles	LinkedIn	Funny	Post	8 / 8 (no rewrite)	121.4	Yes
6	Model failed in production	Instagram	Funny	Reel script	7 / 6	179.3	Partly
7	Final year project	LinkedIn	Professional but exciting	Post	7 / 6	210.3	No
8	Computer vision project	YouTube Shorts	Professional but exciting	Reel script	7 / 7	265.8	Partly
9	Debugging FastAPI	X/Twitter	Funny	Post	7 / 7	112.4	Partly
10	Open-source contribution	LinkedIn	Motivational	Post	7 / 7	178.6	Partly

Tools used in every test: Trend Idea Generator, Caption Writer, Hashtag Generator, Content Reviewer, File Saver and Memory Tool; Reel Script Generator for tests 3, 6 and 8; Self-Improve (rewrite) in 9 of 10 tests. Saved files are in the outputs folder (more than 5 outputs).

Usefulness (my manual review of the saved files): 1 Yes, 7 Partly, 2 No. **Yes:** test 5 (data annotation) - the humor works in places and the post is clear. **Partly:** tests 1, 2, 4, 6, 8, 9, 10 - technically correct but generic, because the agent does not know the user's real experience; the funny tone was often lost in the post body (tests 6, 9). **No:** test 3 (no humor in the post and the reel script is cut off) and test 7 (invented claims such as a 'novel approach' that 'outperforms existing solutions').

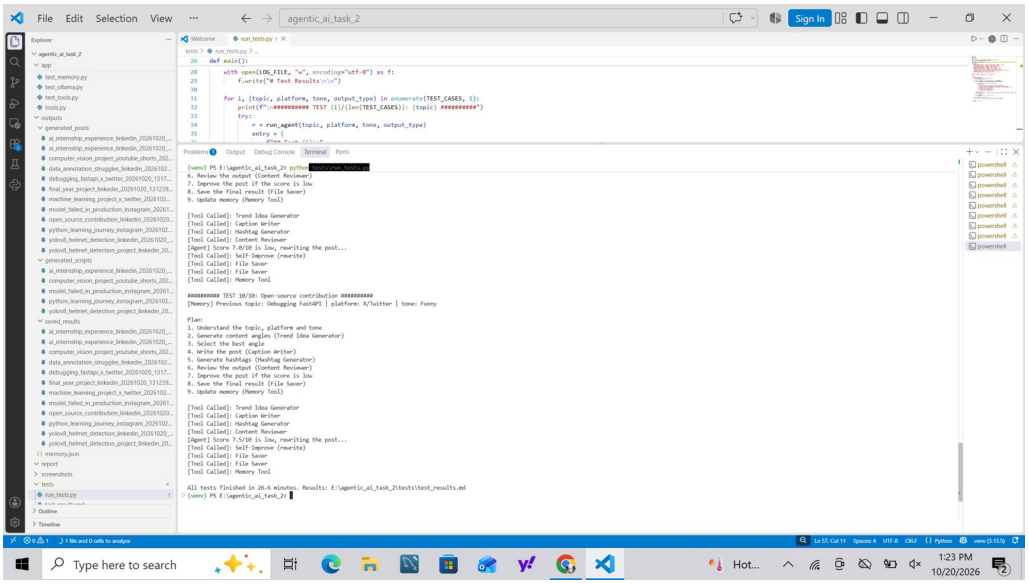


Figure 6: Terminal after the first full 10-test run (26.6 minutes)

7. Error Analysis

Test case / area	Problem found	Possible reason	Fix applied
Model failed in production	'Model' understood as a fashion model	Ambiguous word, no domain context	Added AI/ML context to the system prompt; confirmed in the final run (test 6 is now about ML models)
First YOLOv8 run	Placeholders such as [Company Name] and [Your Website/Link]	Not forbidden in the prompt	Added a no-placeholders rule
Content Reviewer	Review text cut off	Token limit too low	Raised the limit and used a short fixed review format
Hashtags	Broken or repeated hashtags inside the post	No strict format rules	One-word hashtags starting with #, no hashtags in the post
AI internship / project posts	Invented facts (industry, results)	Model fills gaps	Added a 'do not invent facts' rule; test 7 still shows invented claims
Reel scripts (tests 3, 6, 8)	Script cut off in the middle of a sentence	Output token limit too low for long scripts	Not fixed yet (future improvement)
Humor (tests 3, 6, 9)	Funny angle chosen but the final post is not funny	Small model loses the tone while writing the post	Not fixed yet - few-shot humor examples needed
Self-improve (rewrite)	Rewrite did not improve the score	Small model reviewing its own work	Added re-review after rewrite; kept as a known limitation

Key finding: the rewrite step improved the score in 0 of 9 cases. It stayed the same in 6 cases and dropped from 7 to 6 in 3 cases (tests 4, 6 and 7). The model's own review score is also noisy (7/10 in 9 of 10 tests), so it is not used as an accuracy measure.

8. What I Learned

- How to run a local LLM with Ollama and call it from Python.

- The difference between a chatbot and an agent: planning, tool use, memory, review and saving.
- How to connect tools to an LLM and log every tool call.
- Prompt design: platform rules, no placeholders, no invented facts and strict output formats fix many weak outputs.
- How to save agent results and keep simple memory in JSON.
- How to test an agent on 10 inputs, compare versions and analyze failures.
- Limits of small models: slow on CPU, weak at humor, and unreliable at reviewing their own work.

9. Limitations and Future Improvements

- Keep the rewrite only if the review score improves; otherwise keep the original post.
- Code-based validation (character counts, hashtag format) instead of relying on the model's review.
- Raise the output token limit so reel scripts are not cut off.
- Few-shot examples in prompts, especially for funny and meme styles.
- Streamlit interface, web search tool and a user feedback loop.
- SQLite memory, PDF export and a multi-agent workflow (planner, writer, reviewer).
- A larger local model or a GPU for faster and better output.

10. LinkedIn Post

Today, I completed AIRI Team PITB - AI Internship Task 2, where I built an end-to-end Agentic AI assistant using a local LLM through Ollama.

I developed **TrendPilot**, an AI agent that takes a topic and generates content ideas, LinkedIn posts, hashtags, reel scripts and saved Markdown files. It plans its work, calls tools, reviews its own output and remembers previous runs.

Tools used: Python, Ollama, Gemma 3:4B, JSON/Markdown and GitHub.

Key learning: Agentic AI is not only about generating text. It is about planning, using tools, storing useful information, improving outputs and completing tasks in a structured way. I also learned that a small model on CPU is slow and cannot reliably judge its own work.

#AgenticAI #Ollama #LocalLLM #Gemma #Python #ArtificialIntelligence #AIRI #PITB #AIInternship #MachineLearning #GenerativeAI